

ProtoHedge Beyond Simulation: An Audited Historical Evaluation of Interpretable Deep Hedging

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Abstract—Deep Hedging learns sequential hedging decisions directly from market states, frictions, and a risk-sensitive objective, but its neural policies are difficult to inspect. ProtoHedge replaces the final black-box action map with a similarity-weighted dictionary of representative market states. The original study found that this bottleneck was nearly as effective as Deep Hedging in synthetic Black–Scholes and stochastic-volatility environments. We ask whether that conclusion survives historical option data and a path-dependent liability. First, using frozen snapshots of the released TensorFlow source, we reproduce the original ordering: ProtoHedge minus Deep Hedging optimized-certainty-equivalent CVaR@50% utility is -9.10×10^{-4} in the executed Black–Scholes notebook configuration and -1.48×10^{-4} under stochastic volatility. We then construct 34,395 contract-consistent rolling episodes for ten U.S. equities and exchange-traded funds, with chronological train, validation, and 2022–2024 test periods, a fixed listed hedge option within each episode, explicit costs, and bounded inventories. The historical study tests European and arithmetic-average Asian short-call liabilities using three seeds and a validation-selected expanded ProtoHedge family. A post-run audit found that a diagnostic-penalized checkpoint score selected the initialized Deep Hedging policy in 12 of 60 fits. A targeted sensitivity that replaces those affected comparisons using pure validation OCE changes the European conclusion from apparent superiority to near parity. Under this conservative sensitivity, European intervals contain zero, while the Asian tail-selected policy has a panel utility advantage of 0.004015 with a 95% paired temporal-block interval [0.001925, 0.006038]. A one-seed SPY sensitivity finds no detectable difference between the historical PyTorch soft-clipping map and the exact TensorFlow Probability map. Prototype assignments are non-uniform and traceable to historical states. The evidence therefore supports synthetic and historical European near parity, together with a promising path-dependent tail result, but not universal ProtoHedge superiority or superiority over an equally tuned Deep Hedging family.

Index Terms—deep hedging, ProtoHedge, interpretable machine learning, option hedging, sequential decision making, conditional value-at-risk, empirical finance

I. INTRODUCTION

Hedging is a sequential decision problem under uncertainty. At every trading date, a hedger must act using incomplete state information, absorb transaction costs, respect position constraints, and manage a terminal loss distribution rather than only an expected outcome. Deep Hedging addresses this problem by training a policy end-to-end against a monetary utility objective [1]. This flexibility is useful when rebalancing is discrete, frictions matter, or market dynamics are too complex

for a closed-form hedge. It also creates a practical problem: the resulting action is the output of a neural network whose economic rationale can be difficult to audit.

ProtoHedge (ProtoHedge) was introduced as an interpretable alternative [2]. It represents a decision as a convex combination of actions attached to a finite library of representative states, or prototypes. The policy still learns from the same risk-sensitive objective, but its action can be traced to a small set of examples. The original experiments reported near parity with a standard Deep Hedging (Deep Hedging) network in Black–Scholes and stochastic-volatility simulators. That finding is valuable, but it leaves two questions open. First, does the released implementation reproduce the reported synthetic model ordering when its exact utility metric and stored artifacts are used? Second, does the performance–interpretability tradeoff survive historical listed-option paths, changing market regimes, and a path-dependent payoff?

These are not merely requests for a larger benchmark. Historical evaluation changes the scientific problem. Listed contracts have finite lives and cannot be replaced each day without implicitly rolling the hedge. Nearby rolling episodes are strongly dependent. A chronological split is needed to prevent future leakage. The current inventory, rather than only the latest trade, must accrue one-step gains. An Asian payoff additionally requires an observable running-average state. Finally, validation and checkpoint rules become part of the estimand: a policy chosen by a diagnostic-penalized score is not the same policy as one chosen by the paper’s utility.

We address these issues in a sequence of linked experiments, summarized in Fig. 1. We first execute the released TensorFlow model from frozen source snapshots in the two synthetic environments. We then evaluate a PyTorch historical implementation on 34,395 contract-consistent 20-decision episodes spanning ten liquid equities and exchange-traded funds. The test calendar is 2022–2024. All candidate selection is performed on the training and validation periods. We compare unhedged exposure, a market-delta rule, a validation-tuned no-trade band, one fixed paper-style DH architecture, and a validation-selected expanded PH family. We finally audit checkpoint semantics, the scope of the expanded family, the PyTorch SoftClip approximation, and the empirical prototype assignments.

The principal result is deliberately qualified. The original

From implementation fidelity to audited out-of-sample evidence



Fig. 1. Evidence flow. Synthetic confirmation and the historical study answer different questions and are not pooled. The final historical interpretation incorporates targeted checkpoint and implementation-sensitivity audits.

executed historical checkpoint rule suggested a broad PH advantage, especially for European calls. The audit showed that this result was partly attributable to the benchmark checkpoint rule: 12 of 60 DH fits selected their initialized state. Replacing the identified affected comparisons under pure validation OCE reduced the European PH–DH utility gaps by 89–92%, and both European intervals then included zero. This near-parity result is consistent with the original synthetic evidence rather than a failed outcome. For Asian calls, the tail-selected PH result changed little and retained a positive panel interval. Thus, the most defensible interpretation is that the prototype bottleneck preserves performance on terminal-payoff historical tasks and may offer a tail advantage when the liability is path dependent.

The paper makes five contributions:

- 1) We provide a source-level reproduction of the original PH–DH comparison under Black–Scholes and stochastic-volatility dynamics, using the original OCE CVaR@50% utility and documenting a material discrepancy between the Black–Scholes prose specification and executed notebook.
- 2) We construct a leakage-resistant historical panel in which each episode follows one fixed listed option contract, contains 20 trade decisions and 21 marks, and lies wholly inside a chronological period.
- 3) We extend the comparison to European and arithmetic-average Asian liabilities with common information, accounting, costs, bounds, and test paths.
- 4) We use paired temporal-block inference and add direct comparisons with a transparent Spot-Delta baseline, rather than treating DH as the only benchmark.
- 5) We audit checkpoint selection, model-family scope, action clipping, saturation, and prototype usage. The audit changes the headline European claim and is therefore part of the scientific result, not an implementation footnote.

Our claim is correspondingly narrow. We do not claim that PH dominates DH on every asset or metric, that the historical grid is an unchanged copy of the released PH model,

that the one-seed implementation sensitivity replaces a panel experiment, or that the selected PH family beats a comparably tuned DH family. We claim that (i) the original synthetic near-parity ordering can be recovered from the released workflow, (ii) historical European performance is consistent with near parity after a conservative checkpoint sensitivity, and (iii) a validation-tail-selected expanded PH family exhibits a positive Asian OCE CVaR@50% panel result that merits further confirmation.

II. BACKGROUND AND RELATED WORK

A. Deep Hedging as Risk-Sensitive Sequential Control

Let $P_t \in \mathbb{R}^m$ be the vector of hedge-instrument prices at decision time t , h_{t-1} the inventory carried into that decision, and x_t the observable state. A policy proposes a trade increment

$$a_t = \pi_\theta(x_t), \quad h_t = \Pi_{\mathcal{H}}(h_{t-1} + a_t), \quad (1)$$

where $\Pi_{\mathcal{H}}$ enforces the cumulative-position set. For terminal short-liability cash flow $H \leq 0$ and per-unit trading-cost vector κ_t , the premium-excluding liability offset is

$$\mathcal{O}^\pi = H + \sum_{t=0}^{T-1} h_t^\top (P_{t+1} - P_t) - \sum_{t=0}^{T-1} \kappa_t^\top |a_t|. \quad (2)$$

Higher $\mathcal{O}$ means that the hedge offsets more of the common short liability after costs.

The learned objective is the optimized certainty equivalent used in the Deep Hedging codebase:

$$\mathcal{U}_\lambda(\mathcal{O}) = \sup_{y \in \mathbb{R}} \mathbb{E} [(1 + \lambda) \min(\mathcal{O} + y, 0) - y]. \quad (3)$$

For the software’s `cvar` utility, the associated lower-tail probability is $p = (1 + \lambda)^{-1}$. We set $\lambda = 1$, hence $p = 0.5$. The scalar threshold y is learned jointly with the policy. At evaluation we freeze each trained model’s threshold and average its pathwise utility contribution. We refer to this quantity as *frozen-threshold OCE CVaR@50% utility*. It is the metric used for the original-paper reproduction and the primary PH–DH comparison. It is distinct from the separately calculated empirical lower-tail means at 50% and 5% [3].

A standard DH policy implements π_θ as a feed-forward neural network. It is an essential benchmark because it can consume the same state, optimize the same objective, and obey the same constraints as PH. Related work has studied neural hedging under richer market dynamics [4]; our focus is narrower: the behavior of an interpretable prototype bottleneck under historical data and explicit protocol audits.

B. ProtoHedge

Let $\mathcal{P} = \{p_1, \dots, p_K\}$ be fixed representative states. For a standardized state $\tilde{x}_t$, nonnegative feature weights γ , and a learned shared projection f_ϕ , the historical PH implementation forms

$$z_t = f_\phi(\gamma \odot \tilde{x}_t), \quad z_k^p = f_\phi(\gamma \odot p_k), \quad (4)$$

$$d_{t,k} = \|z_t - z_k^p\|_2^2, \quad \alpha_{t,k} = \frac{\exp(-d_{t,k})}{\sum_{j=1}^K \exp(-d_{t,j})}. \quad (5)$$

Each prototype has a learned bounded action $b_k \in \mathbb{R}^m$, and

$$a_t = \sum_{k=1}^K \alpha_{t,k} b_k. \quad (6)$$

Equation (6) makes the local explanation exact with respect to the model: the reported prototype weights and actions are the quantities that generate the decision. This is stronger than attaching an explanation to a separately trained black box, although it does not by itself prove that each prototype has a stable economic interpretation.

The original PH motivation is a performance–interpretability tradeoff rather than universal performance dominance [2]. Fixed anchors preserve the connection between a prototype and an observed state; the projection and prototype actions remain trainable. Our historical grid retains this mechanism but expands the candidate family, as detailed in Section IV-C.

C. Why Historical Validation Is Different

Simulation provides exact control over dynamics and can generate independent paths. Historical option data instead combine non-stationarity, quote noise, liquidity variation, market closures, overlapping trajectories, and a finite set of listed contracts. A valid empirical test must therefore specify the contract path, payoff state, temporal split, dependence-aware uncertainty, and model selection process. These design choices are central to this study. They also explain why the synthetic and historical numbers answer related but non-interchangeable questions: the former tests implementation fidelity under the original worlds; the latter tests out-of-sample behavior under observed paths.

III. ORIGINAL-SOURCE SYNTHETIC CONFIRMATION

A. Protocol

We exported frozen snapshots of the released TensorFlow implementation rather than substituting the PyTorch historical port. The Black–Scholes confirmation uses source commit 983da37765b0; the stochastic-volatility confirmation uses

the archived released source at aedb45060622. Both use TensorFlow 2.13, 20 steps of length 0.02, 10,000 training paths, 1,000 validation paths, 800 epochs, Adam with learning rate 10^{-3} , batch size 32, model seed 1, and the OCE objective in Eq. (3). The DH network has width 20, depth 3, and Softplus activations. PH uses $K = 100$ for Black–Scholes and $K = 500$ for stochastic volatility, following the corresponding released notebook assets. The paired deterministic test contains 10,000 Black–Scholes paths and 5,000 stochastic-volatility paths.

The checkpoint rule matches the original trainer: minimize training OCE over the initialized checkpoint and 800 trained epochs. This rule is retained here because the purpose is faithful reproduction. It should not be confused with the chronological validation selection used for historical data.

B. Black–Scholes Configuration Reconciliation

The released materials contain a reproducibility ambiguity. The prose-level paper configuration indicates zero drift and names a zero-drift, $K = 100$ prototype file, but that file is absent from the released revisions. The executed author notebook instead uses the simulator’s default drift 0.1 and the committed $K = 100$ prototype asset. We report both paths rather than hiding the discrepancy. Regenerating the missing zero-drift prototypes produced a PH–DH gap of -0.008841 . Replaying the executed notebook configuration with its committed prototypes reduced the gap to -0.000910 , while the saved notebook outputs imply -0.000255 . Table I records the reconciliation. The executed-notebook configuration is our headline Black–Scholes confirmation because it is the closest reproducible account of what generated the stored author results; the zero-drift attempt remains part of the archive as a sensitivity.

C. Synthetic Results

Table II and Fig. 2 show the confirmed results. PH is slightly below DH in both environments, reproducing the direction reported in the original paper. The Black–Scholes gap is -0.000910 versus the paper’s approximately -0.000230 ; the stochastic-volatility gap is -0.000148 versus approximately -0.000300 . Absolute Black–Scholes utility errors relative to the saved notebook are 6.3×10^{-5} for DH and 5.92×10^{-4} for PH. Given stochastic optimization and an unrecorded original test-clone seed, this is a sufficiently close directional and scale confirmation for our purpose, but not bitwise replication.

The Black–Scholes DH model has 943 trainable parameters and PH has 102, a $9.25 \times$ ratio. This compactness does not hold mechanically for every K : the $K = 500$ stochastic-volatility PH model has 1,359 parameters versus 1,004 for DH. Parameter efficiency is therefore configuration dependent, while the prototype-based action decomposition remains present by construction.

IV. HISTORICAL EXPERIMENTAL DESIGN

A. Data and Contract-Consistent Episodes

We combine historical U.S. listed-option data from Option-Metrics IvyDB US with daily underlying histories [5], [6].

Protocol	Drift	Prototype input	DH	PH	PH-DH
Paper-specification attempt	0.0	regenerated K=100	-0.058347	-0.067188	-0.008841
Notebook-faithful confirmation	0.1	committed K=100	-0.058102	-0.059013	-0.000910
Saved author notebook	0.1	committed K=100	-0.058166	-0.058421	-0.000255

TABLE I

RECONCILIATION OF THE RELEASED BLACK-SCHOLES SPECIFICATION AND EXECUTED NOTEBOOK. UTILITY IS FROZEN-THRESHOLD OCE CVAR@50%; HIGHER IS BETTER. THE MISSING ZERO-DRIFT PROTOTYPE INPUT CANNOT BE RECOVERED EXACTLY AND WAS REGENERATED IN THE FIRST ROW.

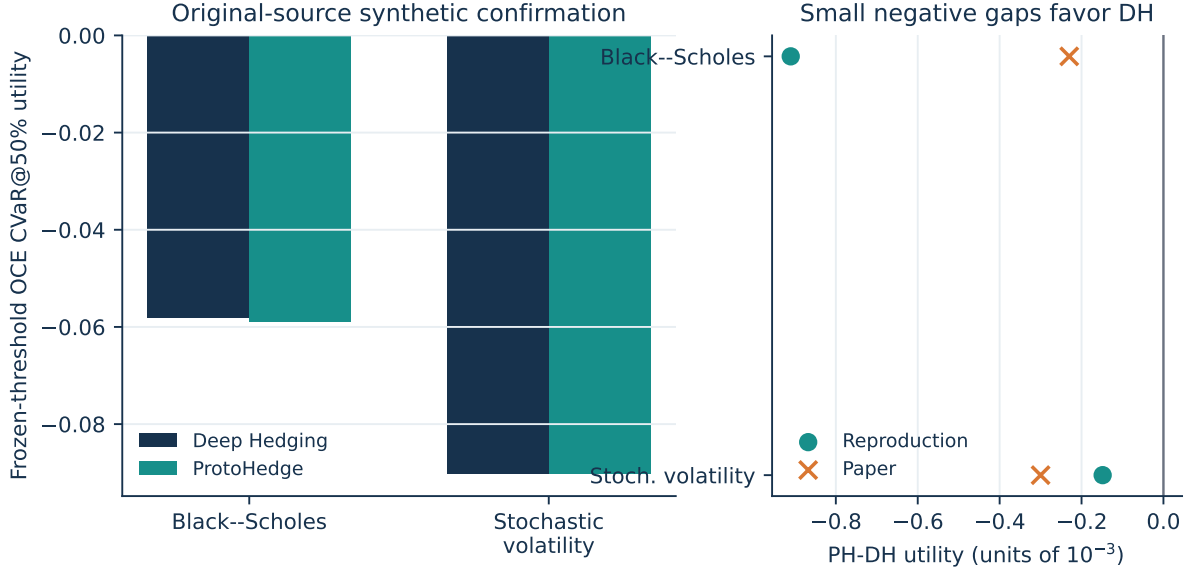


Fig. 2. Synthetic utility levels (left) and PH-DH differences (right). Both confirmations preserve the original ordering and near-parity interpretation.

Environment	Deep Hedging	ProtoHedge	PH-DH	Paper PH-DH
Black-Scholes	-0.058102	-0.059013	-0.000910	-0.000230
Stochastic volatility	-0.090150	-0.090299	-0.000148	-0.000300

TABLE II

ORIGINAL-SOURCE SYNTHETIC CONFIRMATION. PH-DH IS THE DIFFERENCE IN FROZEN-THRESHOLD OCE CVAR@50% UTILITY; NEGATIVE VALUES FAVOR DH. THE FINAL COLUMN REPORTS THE APPROXIMATE DIFFERENCES IN THE ORIGINAL PH PAPER.

The final panel contains AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, and XLV. The underlying set spans single names, broad-market and sector equity funds, and a Treasury fund. The available feature dates run from January 2005 through December 2024.

We retain American-style calls with finite implied volatility and vendor Greeks, positive bids, non-crossed offers, positive midquotes, and positive implied volatility. The listed option price is the bid-ask midpoint. At the start of each candidate episode, eligible contracts have 20–40 calendar days to expiration, initial open interest of at least one, relative spread at most one, and absolute strike moneyness at most 10%. Candidates are ranked by absolute strike distance from spot, distance from a 30-day target maturity, spread, open interest, and volume.

Most importantly, selection occurs once. An episode is

retained only if the same `optionid`, expiration, and strike have valid quotes at all 21 observations and the option expires after the episode. Thus, the listed option is a genuine fixed hedge instrument rather than a daily sequence of unrelated near-ATM contracts. A new contiguous block begins after any gap longer than four calendar days.

The calendar is split before rolling windows are constructed. Training ends on December 31, 2018; validation spans 2019–2021; and test spans 2022–2024. A 20-observation embargo is removed at each post-training boundary. No episode crosses a period, although adjacent episodes within a period can overlap. The corrected panel contains 34,395 episodes: 20,537 training, 6,881 validation, and 6,977 test episodes. Counts by ticker appear in Table III.

The raw feature tensor at each observation is

$$x_t^{\text{raw}} = (S_t, C_t, \Delta_t^{\text{call}}, \nu_t, \sigma_t^{\text{impl}}), \quad (7)$$

where C_t is the fixed listed call midpoint, Δ_t^{call} and ν_t are its vendor delta and vega, and σ_t^{impl} is its implied volatility. Spot, option price, and vega are divided pathwise by S_0 . Policy features are current hedge prices, current cumulative inventory, and time remaining. The Asian policy additionally observes running-average moneyness. In the implementation,

Ticker	Train	Validation	Test	Total
AAPL	2,081	686	711	3,478
IWM	2,352	696	713	3,761
NVDA	1,775	666	644	3,085
QQQ	2,350	713	692	3,755
SPY	2,390	716	713	3,819
TLT	2,026	706	712	3,444
XLE	2,214	640	710	3,564
XLF	1,939	682	705	3,326
XLK	1,643	697	697	3,037
XLV	1,767	679	680	3,126
Total	20,537	6,881	6,977	34,395

TABLE III

CONTRACT-CONSISTENT HISTORICAL EPISODES. EACH ROW CONTAINS 20 DECISION INTERVALS AND 21 OBSERVATIONS. WINDOWS MAY OVERLAP WITHIN A PERIOD BUT CANNOT CROSS THE CHRONOLOGICAL SPLIT OR EMBARGO.

the feature named `delta` is the current inventory h_t ; it should not be confused with the market-provided option sensitivity Δ_t^{call} used by the rule-based baseline.

B. Liabilities, Accounting, and Costs

Each historical episode defines a synthetic liability struck at the initial spot $K_i = S_{i,0}$. The European short call pays

$$H_i^{\text{Euro}} = -\max(S_{i,T} - K_i, 0). \quad (8)$$

The arithmetic-average Asian short call uses the 20 end-of-interval fixings, excluding the initial mark:

$$H_i^{\text{Asian}} = -\max\left(\frac{1}{T} \sum_{t=1}^T S_{i,t} - K_i, 0\right). \quad (9)$$

Before the first fixing, the observable running average is zero; subsequently the policy sees the average of all observed included fixings divided by K_i . Time remaining disambiguates the stage of the averaging window. This feature is provided consistently to DH, PH, prototype extraction, and interpretation.

The hedge trades two instruments: the underlying and the fixed listed call. Actions are trade increments, while gains are earned by the post-trade cumulative inventory. For episode i ,

$$\text{TG}_i = \sum_{t=0}^{T-1} [h_{i,t}^S(S_{i,t+1} - S_{i,t}) + h_{i,t}^C(C_{i,t+1} - C_{i,t})]. \quad (10)$$

This inventory-step accounting prevents positions established at earlier times from disappearing from later gains.

We use the stylized cost model inherited from the synthetic framework:

$$c_{i,t}^S = 0.0002|S_{i,t}|, \quad (11)$$

$$c_{i,t}^C = 0.02|\nu_{i,t}| + 0.0002|\Delta_{i,t}^{\text{call}}| + 0.0005|C_{i,t}|, \quad (12)$$

$$\text{Cost}_i = \sum_{t=0}^{T-1} (|a_{i,t}^S|c_{i,t}^S + |a_{i,t}^C|c_{i,t}^C). \quad (13)$$

These costs are a deterministic friction proxy, not reconstructed executable slippage. The outcome is $\mathcal{O}_i = H_i + \text{TG}_i - \text{Cost}_i$.

The synthetic liability's sale premium is not observed. The midpoint of the listed hedge option is not that premium because the listed option and liability are distinct contracts. Therefore $\mathcal{O}$ is not complete economic profit and loss, and zero is not a break-even threshold. Adding any common exogenous liability premium would shift all strategies for an episode equally, leaving paired mean differences unchanged but potentially changing nonlinear utility levels. We retain the premium-excluding convention for transparent common-task comparison.

C. Models and Candidate Family

We evaluate the following policies.

Unhedged takes no action. **Spot-Delta** holds the observed delta of the selected listed call in spot and zero units of the option. Because the liability itself is synthetic, this is a transparent market-delta proxy rather than its exact analytical delta. **Spot-Delta Band** leaves the current spot inventory unchanged inside $[\Delta_t^{\text{call}} - b, \Delta_t^{\text{call}} + b]$ and projects to the nearest edge outside the band. The width $b \in \{0, .01, .02, .05, .10, .15\}$ is selected by validation mean offset, in the spirit of no-transaction regions [7].

The **Deep Hedging** benchmark is one fixed paper-style architecture per seed, ticker, and liability: width 20, depth 3, Softplus activations, and a two-dimensional trade output. European and Asian policies receive the task-appropriate state described above.

The **expanded ProtoHedge family** uses the same state and output. The shared projection is $D \rightarrow 32 \rightarrow D$ with a ReLU between its linear layers. We evaluate

$$K \in \{10, 25, 50, 100\}, \quad (14)$$

prototypes extracted either from DH trajectories or Spot-Delta trajectories, and unweighted or hand-weighted similarity. The hand-weighted version assigns weight five to time remaining and one to every other state component.

For each source policy, training states are standardized, clustered using K -means [8], and represented by the nearest observed state to each centroid rather than the abstract centroid itself. Anchors and hand-specified feature weights

stay fixed; the projection and prototype action vectors are learned. The released-paper-like subset is the vanilla-DH-derived, unweighted, fixed-feature-weight family. Spot-Delta-derived anchors and hand-weighted similarity are extensions. Of the 40 historical headline mean/tail selections across ticker and liability, 15 belong to the released-paper-like subset and 25 use at least one extension. Consequently, the historical headline must be attributed to the validation-selected *expanded PH family*, not to the unchanged released model.

D. Training and Validation Selection

Neural policies use seeds 1234, 2345, and 3456, an 800-epoch budget, learning rate 10^{-3} , and the OCE CVaR@50% objective. Training adds mild mean absolute-action and inventory penalties with weights 0.001 and 0.002. Trade increments and cumulative positions for both instruments are constrained to $[-1, 1]$. The test calendar is fixed across seeds.

The completed run used a validation checkpoint score

$$\begin{aligned} S = L_{\text{val}} + 0.005 \overline{|a|} + 0.010 \overline{|h|} \\ + 0.100 \text{BO}, \end{aligned} \quad (15)$$

where L_{val} is negative OCE utility and BO is validation bound occupancy. The initialized pre-training state was an eligible checkpoint and is encoded as `selected_epoch=-1`. This value is a real, loadable model, not a missing checkpoint or an 801st epoch.

Within each ticker and liability, candidate-level validation metrics are averaged across seeds. PH candidates must have mean validation bound occupancy no greater than 0.5. Two configurations are then frozen: *Validation-Mean* maximizes validation mean offset, and *Validation-Tail* maximizes validation empirical CVaR@5%. Deterministic tie-breakers use the other metric, lower occupancy, smaller K , and model name. Only those frozen configurations are evaluated on test.

This model-development budget is asymmetric. PH evaluates 16 hyperparameter configurations per seed, while DH uses one architecture. The design is valid for asking whether a validation-selected PH family generalizes against the original fixed DH architecture; it is insufficient for claiming superiority over an optimally tuned neural benchmark.

E. Metrics and Dependence-Aware Inference

The primary cross-model metric is frozen-threshold OCE CVaR@50% utility from Eq. (3). We also report mean offset, empirical CVaR@50%, empirical CVaR@5%, root mean squared offset,

$$\text{RMSE}(\mathcal{O}) = \sqrt{\mathbb{E}[\mathcal{O}^2]}, \quad (16)$$

and downside deviation,

$$\text{DD}(\mathcal{O}) = \sqrt{\mathbb{E}[\min(\mathcal{O}, 0)^2]}. \quad (17)$$

Higher utility, mean, and lower-tail means are better; lower RMSE and downside deviation are better. We also record the fraction of path-time-instrument positions at a cumulative bound and the fraction of paths touching any bound. Because

the premium is excluded, the frequency $\mathcal{O} < 0$ is only a diagnostic and is not interpreted as a shortfall probability.

Adjacent rolling episodes can share most of their dates. We first average the pathwise result across seeds. Within each ticker, we then perform 2,000 paired circular-block bootstrap draws with block length 20 [9]. Panel draws independently resample within ticker, recompute each ticker statistic, and average tickers with equal weight. All differences are oriented so positive values favor PH. Reported 95% intervals are percentile intervals and are not adjusted for the four related headline comparisons; the tail-selected Asian result should therefore be read as strong study-specific evidence rather than a universal hypothesis test.

V. CHECKPOINT AND IMPLEMENTATION AUDITS

A. Meaning and Consequence of `selected_epoch=-1`

The initial checkpoint is allowed by both the original TensorFlow trainer and the PyTorch trainer. Initialization winning is not intrinsically erroneous. It became concerning historically because Eq. (15) mixes the utility objective with action, inventory, and saturation diagnostics. In a constrained hedge, a useful trained policy may appropriately use a boundary; the score can then prefer a weak near-zero initialized policy even when the trained policy has better OCE utility.

The audit found initialization in 12 of 60 DH fits: 11 of 30 European and one of 30 Asian fits. No selected PH fit used initialization. Complete historical epoch trajectories were not serialized, so the old artifacts could not simply be re-ranked after the fact.

We therefore ran a locked targeted sensitivity on the 12 affected DH fits and the two already validation-selected matched PH configurations for each affected ticker/seed, for 36 restart-safe fits in total. The only semantic change was checkpoint selection: unpenalized validation OCE replaced Eq. (15); initialization remained eligible and saturation remained a reported diagnostic. Unaffected historical comparisons were retained. The resulting panel is therefore a *hybrid targeted sensitivity*, not a complete panel-wide rerun under a new selector. We use it as the conservative headline PH-DH estimate because it directly removes the identified initialization failure mode, while explicitly preserving this scope limitation.

B. PyTorch SoftClip Fidelity

The completed historical PH run used a smooth bounded PyTorch map that was not numerically identical to the TensorFlow Probability `SoftClip` used in the released source. Both maps are differentiable and bounded, but calling the full historical implementation an exact numerical port would be too strong. We implemented the original `normalize-to-[0, 1]`, `TFP-softclip`, and `rescale` formula and verified it with six unit tests.

We then trained matched exact-map and historical-map PH policies for European SPY, seed 2345, using a released-paper-like $K = 25$, vanilla-derived, unweighted configuration. The fixed DH comparator and all test paths were held constant.

This is intentionally a small implementation sensitivity, not a replacement for the ten-ticker panel.

VI. HISTORICAL RESULTS

A. The Checkpoint Audit Changes the European Claim

Table IV and Fig. 3 compare the executed protocol with the targeted sensitivity. Under the executed diagnostic-penalized selector, European PH–DH OCE differences were +0.008841 for Validation-Mean and +0.006693 for Validation-Tail, with intervals above zero. After replacing the affected initialized DH checkpoints under pure validation OCE, the estimates fell to +0.001009 and +0.000514. These are 88.6% and 92.3% reductions in absolute gap. Both revised intervals include zero.

The Asian changes are smaller. Validation-Mean falls from +0.003449 to +0.001859, and its revised interval narrowly crosses zero. Validation-Tail changes from +0.004151 to +0.004015, a 3.3% reduction, and its interval remains positive. The asymmetric impact follows the audit incidence: 11 European DH fits but only one Asian DH fit selected initialization.

This audit prevents a stronger but unreliable narrative. The result is not that the historical study failed. It is that European historical near parity is the finding most consistent with both the corrected benchmark and the original synthetic study.

B. Conservative PH–DH Panel Results

Table V and Fig. 4 report the targeted sensitivity. The European Validation-Mean estimate is +0.001009 with interval $[-0.002777, 0.005364]$; Validation-Tail is +0.000514 with $[-0.003418, 0.004734]$. Neither supports a detectable difference.

For Asian calls, Validation-Mean is +0.001859 with $[-0.000309, 0.003863]$, suggestive but not interval-separated from zero. The Validation-Tail estimate is +0.004015 with $[0.001925, 0.006038]$, and every one of the 2,000 panel bootstrap draws is positive. This is the one headline comparison whose interval excludes zero.

Ticker-level results are heterogeneous. The Asian tail estimate is positive for seven of ten tickers, with positive individual intervals for AAPL, NVDA, TLT, and XLF and no ticker interval wholly below zero. European tail estimates are positive for four of ten tickers; only TLT has an interval wholly above zero. For Validation-Mean, Asian estimates are positive for seven tickers but TLT is detectably negative, while European estimates are positive for seven tickers with positive intervals for IWM and TLT. These patterns rule out a uniform dominance claim and motivate the equal-ticker panel summary rather than selective asset reporting.

C. Comparison with Spot-Delta

The Spot-Delta comparisons in Table VI use the completed historical PH checkpoints and are unaffected by replacement of DH checkpoints. For European calls, both PH selections improve mean offset over Spot-Delta: +0.008849 for Validation-Mean and +0.005488 for Validation-Tail, with positive intervals. Their empirical CVaR@50% differences are much smaller and both intervals cross zero. Thus, the European PH

family improves the center of the premium-excluding offset distribution relative to this simple rule without a supported 50% lower-tail advantage.

For Asian calls, both selection rules have positive intervals for both mean offset and empirical CVaR@50%. Validation-Tail improves mean by 0.002795 and empirical CVaR@50% by 0.005830; Validation-Mean improves them by 0.004647 and 0.005139. This reinforces the path-dependent result against a transparent baseline, although it remains tied to the selected expanded PH family and the executed PH checkpoint semantics.

D. Saturation and Descriptive Performance

Under the executed protocol, average DH test bound occupancy is approximately 39% for both liabilities, and 69% of paths touch a bound at least once when pooled. The validation-selected PH rows have much lower average occupancy in the descriptive table: approximately 4.2–5.3% for European calls and 1.6–2.3% for Asian calls. This supports the diagnostic statement that the PH policies do not obtain their executed-protocol performance by occupying the hard bounds more often than DH. It is not a causal explanation of performance, and saturation is not used to define the conservative OCE result.

The full executed-protocol descriptive metrics are retained in Appendix B. They include mean offset, empirical CVaR@5%, RMSE, and occupancy for all six model labels. We do not use those raw model levels to restore the pre-audit European superiority claim.

E. SoftClip Sensitivity

On the matched European SPY check, exact-TFP PH has OCE utility -0.027027 , historical-map PH has -0.029020 , and DH has -0.028771 . Exact PH minus DH is +0.001745 with interval $[-0.009190, 0.014586]$; historical PH minus DH is -0.000249 with $[-0.011394, 0.013623]$. Exact minus historical PH is +0.001993 with $[-0.001693, 0.006412]$. Every interval includes zero. This small check finds no detectable reversal attributable to the map, while also showing that the maps are not numerically interchangeable. The full panel should therefore be described as a close architectural port with an audited action-map sensitivity, not an exact numerical port.

F. Prototype Usage and Example-Based Inspection

We examine QQQ as a representative interpretability case because complete artifacts are available for both liabilities. This analysis uses the PH models selected under the completed historical protocol and is descriptive, not the basis for selecting a performance result.

Let $\bar{\alpha}_k$ be mean assignment mass over test paths and times. Two effective-library sizes are

$$K_{\text{ent}} = \exp \left(- \sum_k \bar{\alpha}_k \log \bar{\alpha}_k \right), \quad K_{\text{inv}} = \left(\sum_k \bar{\alpha}_k^2 \right)^{-1}. \quad (18)$$

Both equal K under uniform use and approach one under concentration.

Liability	Selection	Executed protocol	Targeted sensitivity	Absolute-gap reduction
European	Validation-Mean	+0.008841	+0.001009	+88.6%
European	Validation-Tail	+0.006693	+0.000514	+92.3%
Asian	Validation-Mean	+0.003449	+0.001859	+46.1%
Asian	Validation-Tail	+0.004151	+0.004015	+3.3%

TABLE IV

EFFECT OF THE TARGETED CHECKPOINT SENSITIVITY ON PANEL PH–DH FROZEN-THRESHOLD OCE CVAR@50% DIFFERENCES. “ABSOLUTE-GAP REDUCTION” COMPARES THE MAGNITUDE OF THE CENTRAL ESTIMATE. THE TARGETED COLUMN IS HYBRID BECAUSE UNAFFECTED FITS WERE NOT RERUN.

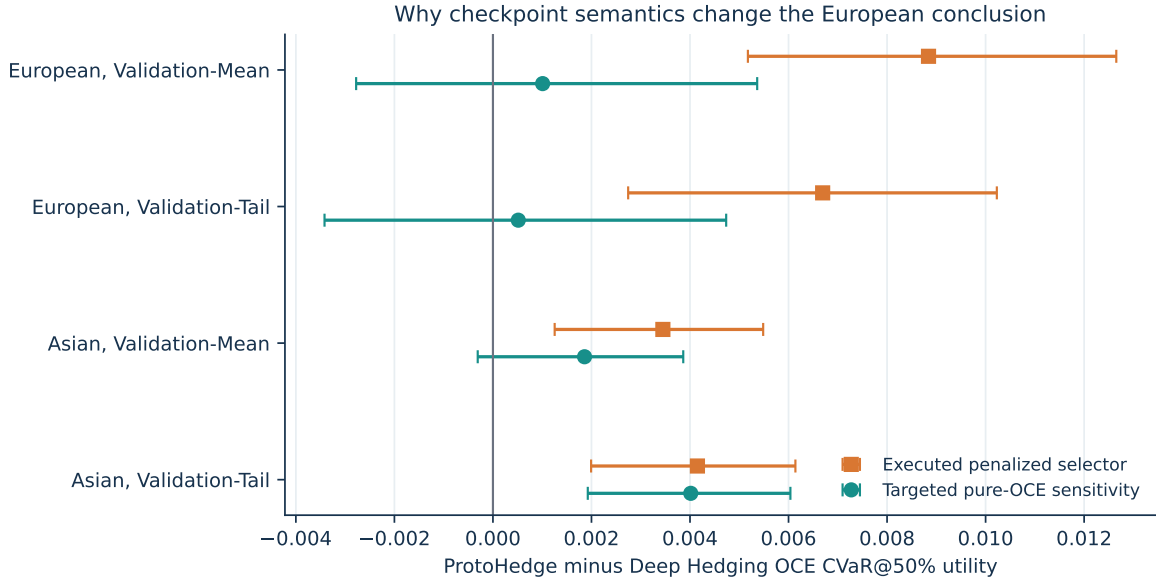


Fig. 3. Executed diagnostic-penalized estimates and targeted pure-OCE sensitivity estimates. The European superiority interpretation does not survive the checkpoint audit; the Asian tail result changes little.

Liability	Selection	PH-DH	95% block interval	$P(\text{PH}_i^c \text{ DH})$
European	Validation-Mean	+0.001009	[-0.002777, +0.005364]	0.696
European	Validation-Tail	+0.000514	[-0.003418, +0.004734]	0.602
Asian	Validation-Mean	+0.001859	[-0.000309, +0.003863]	0.961
Asian	Validation-Tail	+0.004015	[+0.001925, +0.006038]	1.000

TABLE V

CONSERVATIVE HISTORICAL PH–DH DIFFERENCES IN FROZEN-THRESHOLD OCE CVAR@50% UTILITY. POSITIVE VALUES FAVOR THE VALIDATION-SELECTED EXPANDED PH FAMILY. THESE ARE HYBRID TARGETED-SENSITIVITY ESTIMATES, AS DEFINED IN SECTION V.

For European QQ, the nominal $K = 10$ model assigns 24.4% of mass to the most used prototype, 58.2% to the top three, and 83.1% to the top five; its entropy effective size is 7.31. The nominal $K = 50$ Asian selection has mean top-one mass 58.5%, top-ten mass 80.7%, and entropy effective size 14.03. However, the Asian standard deviations are large (0.439 for top-one mass and 20.68 for entropy effective size), revealing strong seed dependence that the mean alone would obscure.

The dominant European prototype changes across return, volatility, and drawdown terciles: P9 dominates down and

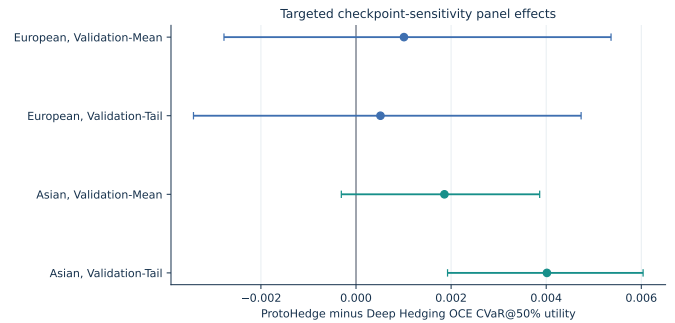


Fig. 4. Panel estimates and paired 95% temporal-block intervals after the targeted checkpoint sensitivity.

middle returns, P2 dominates up returns, P5 dominates low volatility, and P2 dominates middle/high volatility. By contrast, Asian P39 is dominant in every reported bucket, with weight ranging from 0.672 in high volatility to 0.938 in low volatility. Therefore the European case exhibits prototype-identity switching, while the Asian case shows strong concentration with regime-dependent mass but not a changing top prototype.

Liability	Selection	Metric	PH-Spot Delta	95% block interval
Asian	Validation-Mean	Empirical CVaR@50%	+0.005139	[+0.002159, +0.007676]
Asian	Validation-Mean	Mean offset	+0.004647	[+0.002135, +0.007066]
Asian	Validation-Tail	Empirical CVaR@50%	+0.005830	[+0.002925, +0.008393]
Asian	Validation-Tail	Mean offset	+0.002795	[+0.000271, +0.005291]
European	Validation-Mean	Empirical CVaR@50%	+0.001114	[-0.000986, +0.003142]
European	Validation-Mean	Mean offset	+0.008849	[+0.007326, +0.010442]
European	Validation-Tail	Empirical CVaR@50%	+0.000163	[-0.001925, +0.002134]
European	Validation-Tail	Mean offset	+0.005488	[+0.004086, +0.006858]

TABLE VI

PAIRED HISTORICAL COMPARISONS WITH SPOT-DELTA. HIGHER MEAN OFFSET AND EMPIRICAL CVAR@50% ARE BETTER; POSITIVE DIFFERENCES FAVOR PH. THESE ROWS USE THE COMPLETED HISTORICAL PH PROTOCOL AND ARE SEPARATE FROM THE DH CHECKPOINT SENSITIVITY.

Liability	Nominal K	Top-1 mass	Top-3 mass	Top-5 mass	Entropy effective K
European	10	0.244	0.582	0.831	7.31
Asian	50	0.585	0.696	0.738	14.03

TABLE VII

MEAN PROTOTYPE CONCENTRATION FOR THE SELECTED QQQ POLICIES ACROSS THREE SEEDS. THE ASIAN AVERAGES HAVE SUBSTANTIAL CROSS-SEED VARIABILITY AND SHOULD NOT BE INTERPRETED AS A STABLE SINGLE-POLICY SPARSITY ESTIMATE.

The archived nearest-state records connect dominant anchors to concrete test episodes, dates, contract identifiers, inventories, and normalized prices. This supports exact example retrieval. It does not establish that users will find the examples sufficient for every governance task, nor that the prototype identity is stable under retraining.

VII. DISCUSSION

A. Relation to the Original ProtoHedge Result

The original PH paper found a small DH advantage in both synthetic worlds. Our frozen-source experiments recover that direction. The checkpoint-corrected historical European result is also near zero. From a validation perspective, this is arguably more coherent than the initial large historical superiority: the same prototype bottleneck that costs little utility in simulation also costs no detectable utility on the ten-ticker terminal-call panel.

The Asian tail result is different. It is not a reproduction of either synthetic world; it is an extension to a payoff that depends on the path. The policy observes running-average moneyiness, so PH and DH receive payoff-sufficient information. The positive tail-selected panel estimate suggests that the prototype bottleneck may regularize decisions usefully in this setting. The result remains a family-level, validation-selected comparison, however, and the present study cannot distinguish architectural regularization from the benefit of the larger PH search budget.

B. What Is Good About the Result

Four aspects are encouraging. First, the original-source synthetic ordering is recovered under both environments, anchoring the empirical study to the metric used in the original

work. Second, the corrected historical protocol spans ten underlyings and a genuinely held-out three-year test calendar. Third, European near parity survives a conservative benchmark repair, which supports the basic performance–interpretability premise without relying on an implausibly weak DH checkpoint. Fourth, the Asian tail estimate is positive both against DH under the targeted sensitivity and against Spot-Delta under direct block-bootstrap comparison.

The audit itself is also a strength. Reporting the initial stronger claim would have produced a simpler story, but the apparent European advantage was not stable to a scientifically relevant checkpoint rule. Making that change visible improves the reliability and reproducibility of the final result.

C. What Is Weaker Than Initially Expected

The study no longer supports broad European PH superiority. The Asian Validation-Mean interval also crosses zero. Ticker effects are heterogeneous, and the Asian interpretability statistics vary substantially across seeds. The full historical PH family includes extensions absent from the released model, and its development budget is larger than DH’s. Finally, the exact SoftClip check covers one ticker, one liability, one seed, and one PH configuration. These facts narrow the paper from a universal performance claim to a careful empirical validation plus a promising path-dependent result.

D. Implications for Reliable Sequential Decision Making

The experiments illustrate a broader point. Reliability in sequential learning is not only a property of the policy class. It also depends on the data unit, state sufficiency, selection functional, benchmark budget, and uncertainty estimator. Here, a single checkpoint convention altered the European conclusion while leaving the raw test data untouched. Conversely, making the Asian payoff state observable was necessary before any architectural comparison could be meaningful.

Prototype policies offer one useful structural constraint: actions are routed through a finite reference set that can be inspected and retrieved. A natural next theoretical direction is to characterize when such a bottleneck improves robustness under regime shift or limited data. Possible questions include generalization bounds that depend on effective prototype usage

rather than nominal K , stability guarantees for prototype assignments under distribution shift, and risk-sensitive regret bounds for prototype-constrained policies. These questions go beyond the empirical claims of this paper but follow directly from the observed near-parity and path-dependent behavior.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Expanded PH family and unequal tuning. PH searches 16 configurations per seed and DH uses one fixed architecture. The experiment validates the expanded PH family against the paper-style benchmark, not against a hyperparameter-matched DH family. Twenty-five of 40 headline PH selections use at least one extension.

Hybrid checkpoint sensitivity. The pure-OCE rerun is targeted at the 12 identified initialized DH fits and their matched selected PH candidates. It is not a complete rerun of every epoch and every PH configuration under one uniform new selector. We therefore report both executed and sensitivity results and avoid calling the latter a definitive protocol replacement.

Action-map scope. The ten-ticker run uses the historical smooth map; the exact TFP formula is tested only on one European SPY seed. The interval is wide enough that a modest effect cannot be excluded.

Premium and execution. The outcome excludes the liability premium and is not economic P&L. Option prices are daily midquotes, and costs are stylized rather than trade-level realized slippage. The selected listed option’s vendor delta is only a proxy for the delta of the synthetic liability.

Dependence. Circular blocks address overlap within ticker, but the panel resampling does not fully model synchronous cross-ticker market shocks. The block length is fixed at the episode horizon, and no block-length sensitivity is used in the headline table.

Coverage and selection. Ten liquid underlyings do not represent all asset classes, liquidity regimes, strikes, maturities, or option styles. The liabilities are at-the-money, the horizon is fixed, and the validation and test calendars contain specific market regimes. Two PH selection objectives create related comparisons, and intervals are not multiplicity-adjusted.

Synthetic reproducibility. The original Black–Scholes zero-drift prototype artifact is missing, the original test clone seed was not recorded, and neural optimization is stochastic. The notebook-faithful run is close but not numerically identical to the saved output.

Training provenance. The completed historical runner did not embed a Git commit or full package freeze in each raw artifact. The archive records this gap honestly: source and input files were checksummed at archive time, and the locked cloud requirements are retained, but archive-time package versions are not relabeled as training-time versions.

Interpretability. Equation (6) makes the reported weights faithful to action generation, but concentration and nearest examples do not establish causal economic explanations, user usefulness, or retraining stability.

IX. REPRODUCIBILITY AND ARTIFACT INTEGRITY

The completed historical run contains 20 ticker–liability tasks and 4,025 raw files totaling approximately 132 MB. SHA-256 manifests cover raw outputs, 53 panel inputs totaling approximately 136 MB, and 73 archive-time source files. The synthetic directories separately retain source snapshots, environments, protocol JSON files, training histories, model artifacts, test paths, and checksums. The SPY confirmation retains 23 checked artifacts, and the notebook-faithful Black–Scholes confirmation retains 16.

For this manuscript, a deterministic generator copies 43 compact source files into a paper-facing archive, records their original paths and hashes, and creates every CSV, LaTeX table, PDF figure, PNG preview, and a locked numerical results ledger. Generated assets have a second checksum manifest. The generator records Git HEAD 47c80a2b210d, Python 3.12.2, NumPy 1.26.4, pandas 2.2.3, and Matplotlib 3.9.2 for the paper-asset build. These versions describe asset generation, not the historical GPU training environment.

The intended reproduction commands are

```
cd paper/final_manuscript
python generate_assets.py
latexmk -pdf \
    protohedge_historical_validation.tex
```

All manuscript numbers should be edited through the source data or generator, not by changing the rendered tables manually.

X. CONCLUSION

We evaluated whether ProtoHedge’s interpretable action bottleneck survives the transition from stylized simulation to historical option paths. Frozen-source synthetic experiments reproduce the original direction: PH is slightly below DH in both Black–Scholes and stochastic-volatility environments. A corrected ten-underlying historical panel then shows European near parity after a targeted checkpoint repair, aligning the empirical result with that synthetic premise. For arithmetic-average Asian liabilities, the tail-selected expanded PH family retains a positive OCE CVaR@50% panel interval and improves both mean and empirical CVaR@50% relative to Spot-Delta.

The most important methodological finding is that checkpoint semantics matter. The initially stronger European result shrank by roughly 90% after replacing initialized DH selections under pure validation OCE. The final paper is stronger for reporting that correction. The evidence supports PH as an inspectable, competitive historical hedge and identifies path-dependent tail risk as a promising setting, while leaving universal superiority, matched model-family tuning, and full exact-map confirmation to future work.

APPENDIX A

LOCKED HISTORICAL PROTOCOL

Table VIII lists the principal locked values. A machine-readable JSON record is included in the artifact archive.

Component	Locked value
Underlyings	AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, XLV
Liabilities	European and arithmetic-average Asian ATM short calls
Episode timing	20 decisions, 21 observations
Train / validation / test	2005–2018 / 2019–2021 / 2022–2024
Boundary embargo	20 observations
Training seeds	1234, 2345, 3456
Epochs / learning rate	800 / 0.001
DH architecture	width 20, depth 3, Softplus
Prototype counts	10, 25, 50, 100
Prototype sources	Deep Hedging and Spot-Delta
Similarity variants	unweighted and hand-weighted
Trade / position bounds	[-1,1] per instrument
Bootstrap	2000 paired circular-block draws; block 20

TABLE VIII

LOCKED HISTORICAL PROTOCOL. A NULL BATCH OVERRIDE MEANS THE RUNNER’S IMPLEMENTATION DEFAULT WAS RETAINED RATHER THAN RETROACTIVELY ASSIGNING A BATCH SIZE.

APPENDIX B

ORIGINAL EXECUTED-PROTOCOL DESCRIPTIVES

Table IX reports model levels from the completed run. These values are useful for scale, friction, and boundary diagnostics. They are not substituted for the targeted PH–DH OCE sensitivity in the headline claim.

APPENDIX C

PER-TICKER TARGETED SENSITIVITY

Tables X and XI report every ticker; Fig. 5 displays the same intervals. No assets are omitted from panel conclusions.

APPENDIX D

CHECKPOINT AND FAMILY-SCOPE DIAGNOSTICS

APPENDIX E

BLACK–SCHOLES ALIGNMENT DETAIL

APPENDIX F

SOFTCLIP CONFIRMATION DETAIL

APPENDIX G

INTERPRETABILITY DETAIL

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Liability	Model	Mean offset	CVaR@5%	RMSE	Bound occ.
European	Unhedged	-0.031696	-0.137725	0.051082	0.0%
European	Spot-Delta	-0.030281	-0.070542	0.033481	0.0%
European	Spot-Delta Band	-0.029875	-0.073046	0.033661	0.0%
European	Deep Hedging	-0.023872	-0.113276	0.041646	39.0%
European	ProtoHedge (Validation-Mean)	-0.021432	-0.094189	0.035060	5.3%
European	ProtoHedge (Validation-Tail)	-0.024793	-0.084112	0.033853	4.2%
Asian	Unhedged	-0.018544	-0.082546	0.030284	0.0%
Asian	Spot-Delta	-0.017129	-0.065298	0.029230	0.0%
Asian	Spot-Delta Band	-0.016724	-0.066718	0.028730	0.0%
Asian	Deep Hedging	-0.016210	-0.082270	0.030993	39.3%
Asian	ProtoHedge (Validation-Mean)	-0.012483	-0.085456	0.029431	2.3%
Asian	ProtoHedge (Validation-Tail)	-0.014334	-0.077504	0.026745	1.6%

TABLE IX

EQUAL-TICKER AVERAGES UNDER THE ORIGINALLY EXECUTED HISTORICAL CHECKPOINT PROTOCOL. CVaR@5% IS THE EMPIRICAL LOWER 5% MEAN, NOT THE OCE CVaR@50% TRAINING UTILITY. LOWER RMSE AND OCCUPANCY ARE BETTER.

Ticker	Mean sel.	95% interval	Tail sel.	95% interval
AAPL	+0.000790	[-0.0129, +0.0152]	+0.004558	[-0.0092, +0.0204]
IWM	+0.009644	[+0.0033, +0.0166]	+0.001921	[-0.0059, +0.0104]
NVDA	-0.014811	[-0.0407, +0.0167]	-0.012413	[-0.0344, +0.0139]
QQQ	-0.001108	[-0.0096, +0.0086]	-0.000598	[-0.0160, +0.0166]
SPY	+0.000893	[-0.0083, +0.0109]	-0.001670	[-0.0116, +0.0103]
TLT	+0.009319	[+0.0028, +0.0164]	+0.017408	[+0.0083, +0.0267]
XLE	+0.000701	[-0.0079, +0.0105]	-0.000484	[-0.0069, +0.0069]
XLF	+0.004714	[-0.0008, +0.0109]	-0.001828	[-0.0072, +0.0036]
XLK	-0.000783	[-0.0117, +0.0110]	+0.001573	[-0.0053, +0.0092]
XLV	+0.000725	[-0.0069, +0.0082]	-0.003326	[-0.0103, +0.0042]

TABLE X

EUROPEAN PH-DH FROZEN-THRESHOLD OCE CVaR@50% DIFFERENCES UNDER THE HYBRID TARGETED CHECKPOINT SENSITIVITY.

Ticker	Mean sel.	95% interval	Tail sel.	95% interval
AAPL	+0.001200	[-0.0028, +0.0056]	+0.006108	[+0.0016, +0.0110]
IWM	+0.007285	[+0.0011, +0.0129]	+0.004657	[-0.0030, +0.0126]
NVDA	+0.014533	[+0.0062, +0.0220]	+0.014533	[+0.0067, +0.0220]
QQQ	+0.003307	[-0.0019, +0.0082]	+0.003307	[-0.0022, +0.0083]
SPY	+0.001395	[-0.0032, +0.0067]	-0.000483	[-0.0052, +0.0050]
TLT	-0.008643	[-0.0132, -0.0039]	+0.008796	[+0.0053, +0.0127]
XLE	-0.002502	[-0.0141, +0.0076]	-0.001143	[-0.0104, +0.0082]
XLF	+0.003464	[+0.0002, +0.0065]	+0.003464	[+0.0003, +0.0068]
XLK	-0.002604	[-0.0123, +0.0070]	-0.002802	[-0.0122, +0.0067]
XLV	+0.001158	[-0.0039, +0.0060]	+0.003715	[-0.0009, +0.0083]

TABLE XI

ASIAN PH-DH FROZEN-THRESHOLD OCE CVaR@50% DIFFERENCES UNDER THE HYBRID TARGETED CHECKPOINT SENSITIVITY.

Ticker-level checkpoint-audited OCE CVaR@50% effects

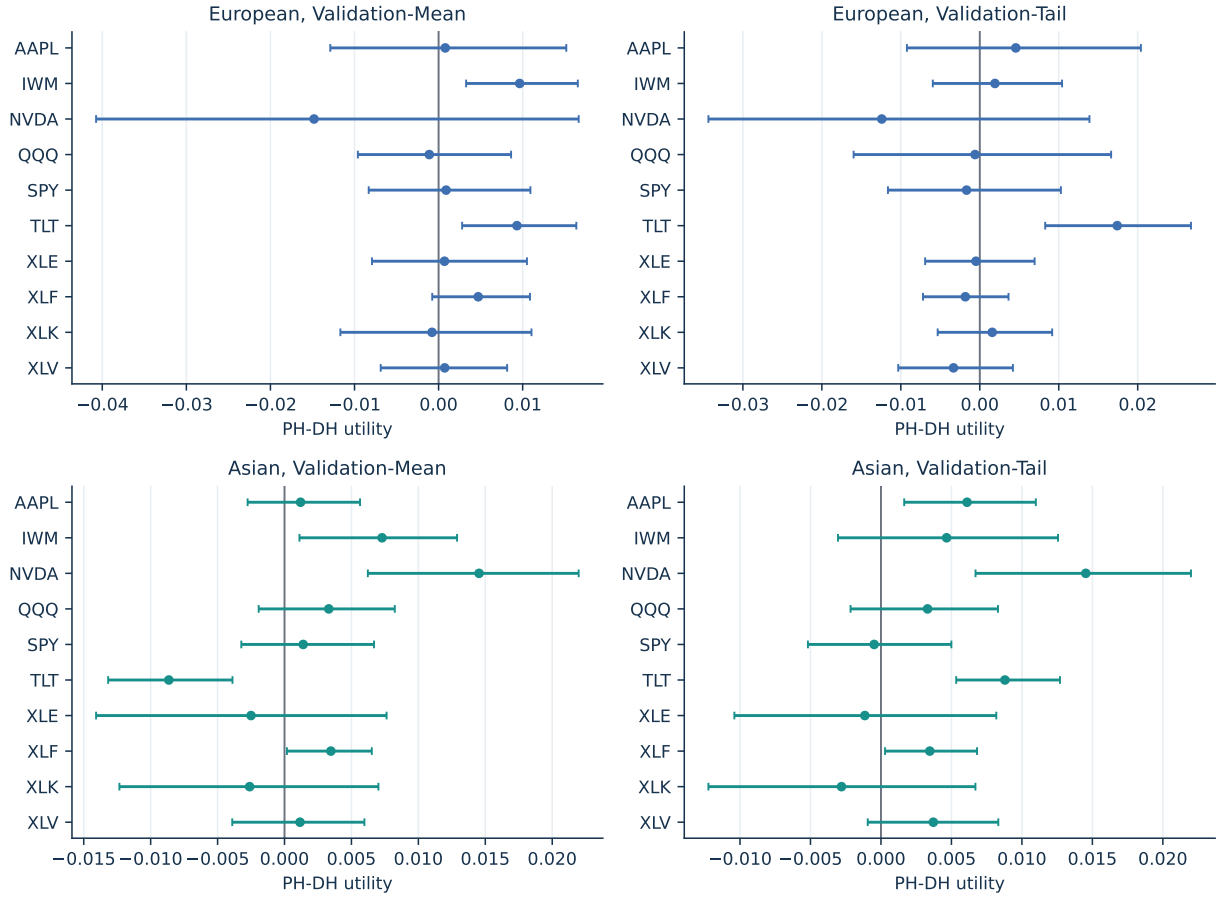


Fig. 5. Ticker-level central estimates and paired temporal-block intervals. Heterogeneity is visible for both liabilities and selection objectives.

Liability	Fits	Epoch -1	Fraction	Median trained epoch	Test bound occ.	Paths touching bound
All	60	12	20.0%	38.0	39.2%	69.1%
Asian	30	1	3.3%	70.0	39.3%	71.7%
European	30	11	36.7%	16.0	39.0%	66.5%

TABLE XII

DEEP HEDGING CHECKPOINT AND SATURATION AUDIT FROM THE COMPLETED RUN. EPOCH -1 DENOTES THE INITIALIZED CHECKPOINT. “PATHS TOUCHING BOUND” IS A PATH-LEVEL DIAGNOSTIC AND CAN BE HIGH EVEN WHEN AVERAGE OCCUPANCY IS LOWER.

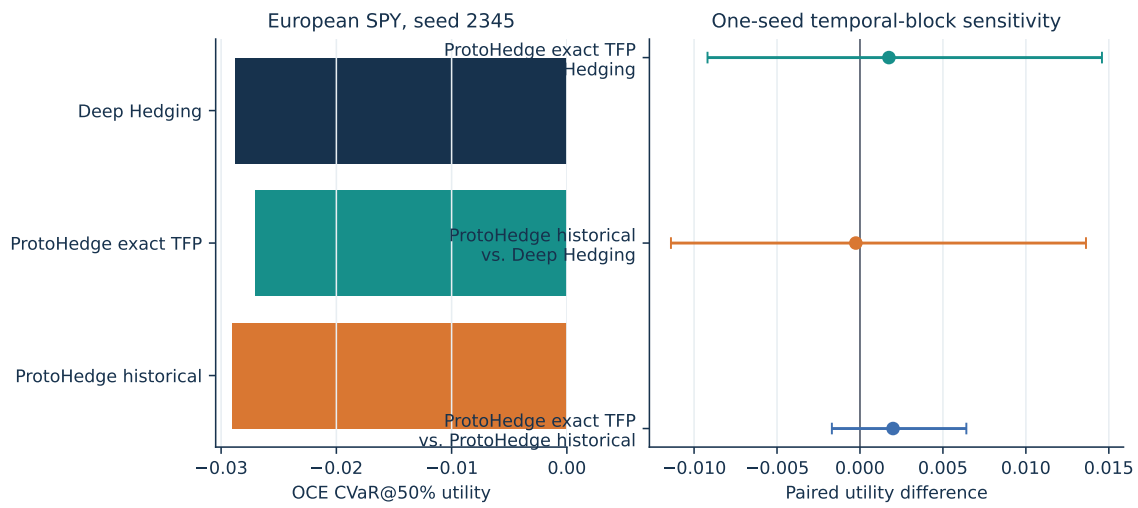


Fig. 6. Model levels and paired contrasts in the one-seed exact-SoftClip SPY confirmation.

Liability	Released-paper-like	Uses extension	Total selections
European	5	15	20
Asian	10	10	20
All	15	25	40

TABLE XIII

SCOPE OF THE SELECTED HISTORICAL PH FAMILY. EACH TICKER CONTRIBUTES A MEAN AND TAIL SELECTION PER LIABILITY.

Model	Reproduction	Saved notebook	Absolute error
Deep Hedging	-0.058102	-0.058166	0.000063
ProtoHedge	-0.059013	-0.058421	0.000592

TABLE XIV

NOTEBOOK-FAITHFUL BLACK-SCHOLES UTILITY LEVELS VERSUS THE VALUES STORED IN THE RELEASED AUTHOR NOTEBOOK.

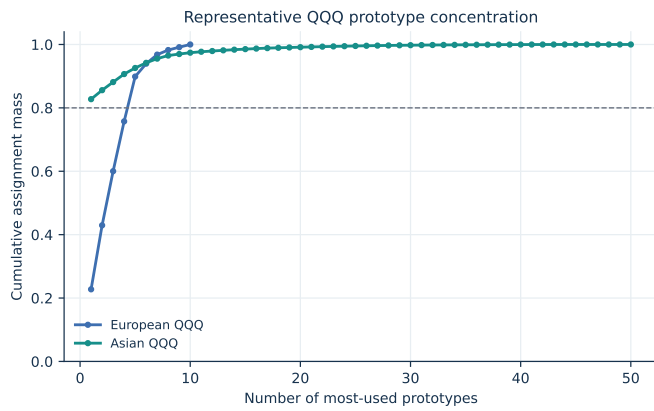


Fig. 7. Cumulative mean prototype assignment mass for representative QQQ policies. The three-seed Asian average hides substantial seed variability.

Model	Epoch	OCE CVaR@50%	Mean	Empirical CVaR@50%	CVaR@5%	Bound occ.
Deep Hedging	180	-0.028771	-0.014452	-0.028549	-0.093524	0.500
ProtoHedge exact TFP	532	-0.027027	-0.015661	-0.025312	-0.056798	0.170
ProtoHedge historical	356	-0.029020	-0.020097	-0.028398	-0.053125	0.073

TABLE XV

EUROPEAN SPY, SEED 2345, MODEL-LEVEL METRICS. THESE ARE A ONE-SEED IMPLEMENTATION SENSITIVITY AND NOT PANEL INFERENCE.

Contrast	Difference	95% block interval	P(≤ 0)
ProtoHedge exact TFP vs. Deep Hedging	+0.001745	[-0.009190, +0.014586]	0.589
ProtoHedge historical vs. Deep Hedging	-0.000249	[-0.011394, +0.013623]	0.471
ProtoHedge exact TFP vs. ProtoHedge historical	+0.001993	[-0.001693, +0.006412]	0.846

TABLE XVI

PAIRED TEMPORAL-BLOCK CONTRASTS FOR THE SOFTCLIP SENSITIVITY. ALL 95% INTERVALS CONTAIN ZERO.

Liability	Partition	Regime	Dominant prototype	Mean weight
European	Drawdown	Deep	P2	0.293
European	Drawdown	Middle	P2	0.260
European	Drawdown	Mild	P9	0.291
European	Return	Down	P9	0.316
European	Return	Middle	P9	0.211
European	Return	Up	P2	0.374
European	Volatility	High	P2	0.249
European	Volatility	Low	P5	0.232
European	Volatility	Middle	P2	0.277
Asian	Drawdown	Deep	P39	0.923
Asian	Drawdown	Middle	P39	0.846
Asian	Drawdown	Mild	P39	0.714
Asian	Return	Down	P39	0.745
Asian	Return	Middle	P39	0.894
Asian	Return	Up	P39	0.844
Asian	Volatility	High	P39	0.672
Asian	Volatility	Low	P39	0.938
Asian	Volatility	Middle	P39	0.872

TABLE XVII

DOMINANT QQQ PROTOTYPE BY RETURN, VOLATILITY, AND DRAWDOWN TERCILE FOR THE COMPLETED-PROTOCOL REPRESENTATIVE POLICIES.

Liability	Proto.	Episode	Step	h^S	h^C	S/S_0	C/S_0	Asian state
European	P1	2024-04-29 to 2024-05-28	6	0.686	0.007	1.017	0.024	–
European	P2	2023-05-25 to 2023-06-26	5	0.889	-0.001	1.044	0.041	–
European	P5	2024-06-14 to 2024-07-16	2	0.571	0.003	1.013	0.019	–
European	P9	2024-05-29 to 2024-06-27	1	0.348	0.003	0.989	0.006	–
Asian	P20	2022-05-24 to 2022-06-23	14	-0.018	0.009	0.961	0.010	1.044
Asian	P27	2022-05-26 to 2022-06-27	11	-0.125	-0.154	0.920	0.006	1.012
Asian	P36	2022-05-24 to 2022-06-23	14	-0.018	0.009	0.961	0.010	1.044
Asian	P39	2022-05-11 to 2022-06-09	5	0.013	-0.137	0.998	0.041	1.020

TABLE XVIII

NEAREST RANK-ONE HISTORICAL STATE FOR SELECTED DOMINANT QQQ PROTOTYPES. h^S AND h^C ARE INVENTORIES; PRICES ARE NORMALIZED BY INITIAL SPOT. THE ASIAN STATE IS RUNNING-AVERAGE MONEYNESS.

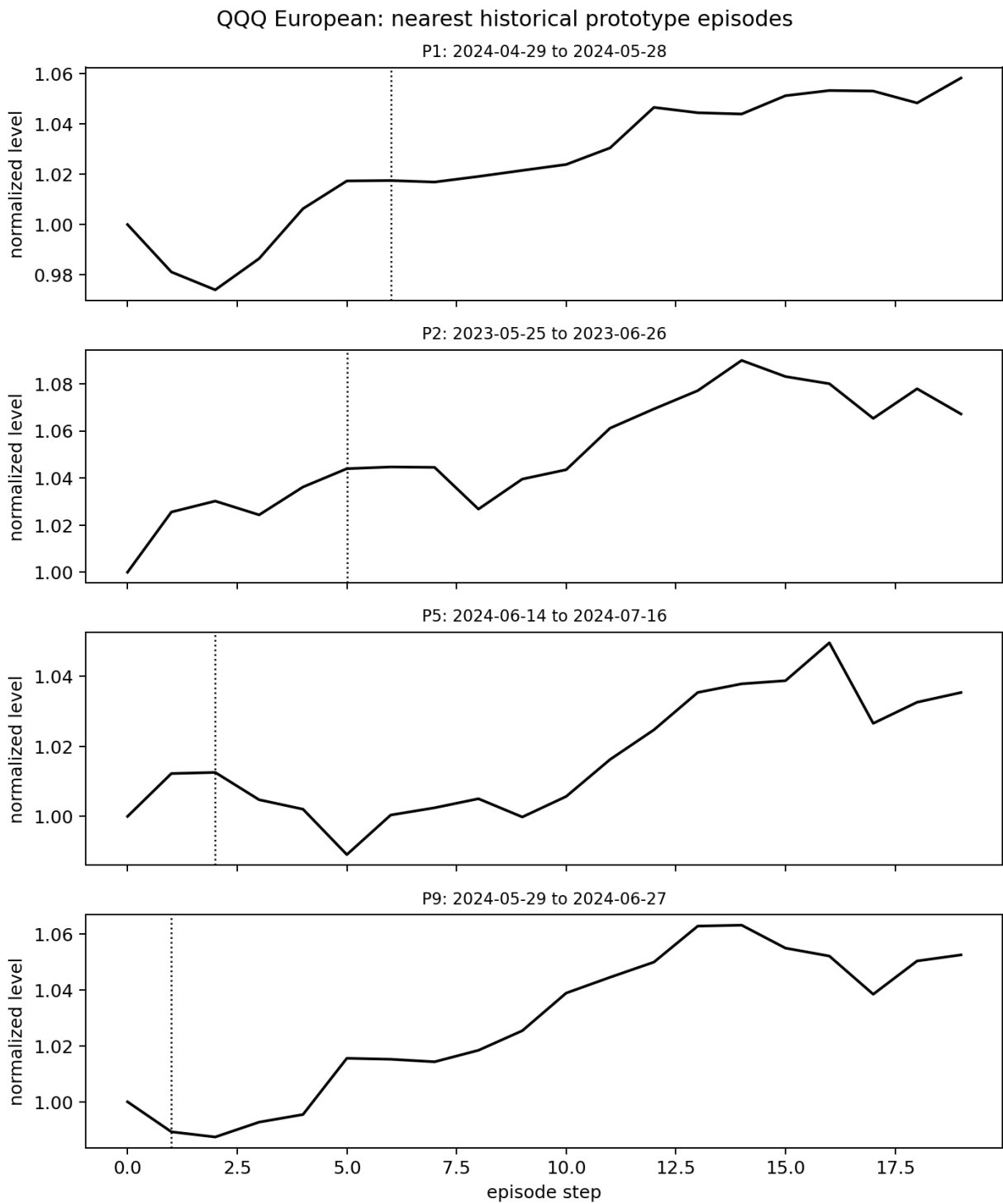


Fig. 8. Retrieved European QQQ historical episodes nearest to selected prototype states. This is an example-based inspection aid, not a performance comparison.

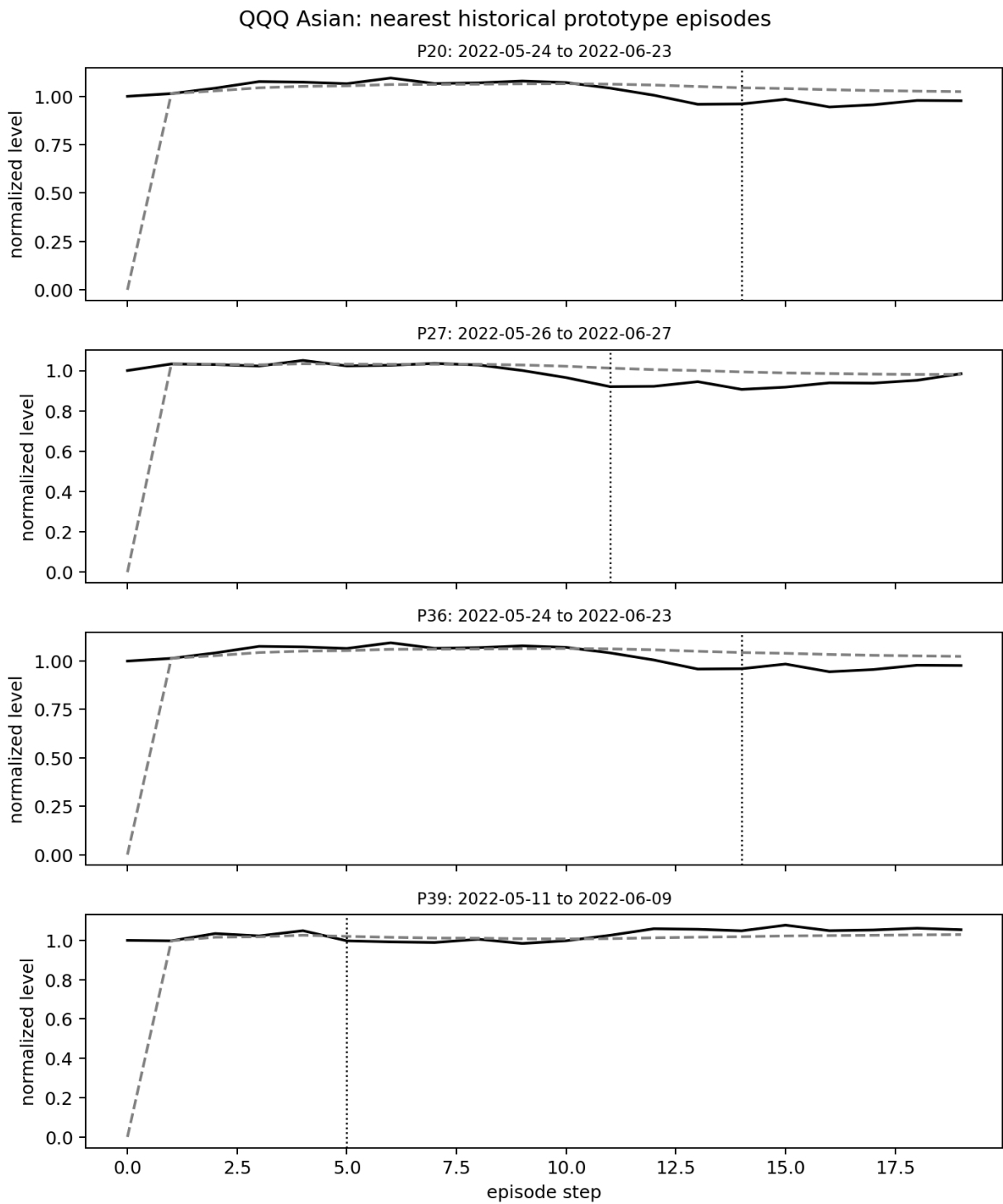


Fig. 9. Retrieved Asian QQQ historical episodes nearest to selected prototype states.